

Problem 2

Part a)

Since $g(0) = 0$ and g is continuously differentiable, the fundamental theorem of calculus gives

$$g(x) = \int_0^x g'(t) dt = \int_0^{+\infty} g'(t) \mathbf{1}_{[0,x]}(t) dt \quad \text{for all } x \geq 0. \quad (1)$$

By the substitution formula (Theorem 3.3.12),

$$\mathbb{E}[g(X)] = \int_{\mathbb{R}} g(x) \mathbb{P}_X(dx).$$

Since $X \geq 0$, the measure $\mathbb{P}_X$ is supported on $[0, +\infty)$, so

$$\mathbb{E}[g(X)] = \int_{[0, +\infty)} g(x) \mathbb{P}_X(dx).$$

Substituting the integral representation of $g(x)$ in (1) gives

$$\mathbb{E}[g(X)] = \int_{[0, +\infty)} \left(\int_0^{+\infty} g'(t) \mathbf{1}_{[0,x]}(t) dt \right) \mathbb{P}_X(dx).$$

The integrand $g'(t) \mathbf{1}_{[0,x]}(t)$ is non-negative since $g' \geq 0$ (as g is monotonically increasing) and $\mathbf{1}_{[0,x]}(t) \geq 0$. Then, by Tonelli's theorem,

$$\begin{aligned} \mathbb{E}[g(X)] &= \int_0^{+\infty} \int_{[0, +\infty)} g'(t) \mathbf{1}_{[0,x]}(t) \mathbb{P}_X(dx) dt \\ &= \int_0^{+\infty} g'(t) \int_{[0, +\infty)} \mathbf{1}_{[0,x]}(t) \mathbb{P}_X(dx) dt. \end{aligned} \quad (2)$$

For a fixed $t \geq 0$, we have $\mathbf{1}_{[0,x]}(t) = 1$ if and only if $x \geq t$, so

$$\int_{[0, +\infty)} \mathbf{1}_{[0,x]}(t) \mathbb{P}_X(dx) = \mathbb{P}_X([t, +\infty)) = \mathbb{P}(X \geq t).$$

Substituting this back into (2), we obtain

$$\mathbb{E}[g(X)] = \int_0^{+\infty} g'(t) \mathbb{P}(X \geq t) dt.$$

□

Part b)

Since Y is $\mathbb{Z}$ -valued, it is a discrete random variable. We assume that the expected value exists, and use Definition 3.3.7:

$$\mathbb{E}[Y] = \mathbb{E}[Y^+] - \mathbb{E}[Y^-], \quad (\text{b.1})$$

where $Y^+(\omega) = \max\{Y(\omega), 0\}$ and $Y^-(\omega) = -\min\{Y(\omega), 0\}$. For the non-negative integer-valued part Y^+ , we can write it as an infinite sum of indicator functions:

$$Y^+ = \sum_{k=1}^{\infty} \mathbf{1}_{\{Y \geq k\}} = \sum_{k=0}^{\infty} \mathbf{1}_{\{Y > k\}}$$

For all $n \in \mathbb{N}$, let $X_n = \sum_{k=0}^n \mathbf{1}_{\{Y > k\}}$. Since indicator functions are non-negative, $(X_n)_{n \in \mathbb{N}}$ is a monotonically increasing sequence of non-negative random variables that converges pointwise to Y^+ , so that we can apply the Monotone Convergence Theorem (Theorem 3.3.11a):

$$\begin{aligned}
\mathbb{E}[Y^+] &= \mathbb{E} \left[\lim_{n \rightarrow \infty} \sum_{k=0}^n \mathbf{1}_{\{Y > k\}} \right] \\
&\stackrel{\text{MCT}}{=} \lim_{n \rightarrow \infty} \mathbb{E} \left[\sum_{k=0}^n \mathbf{1}_{\{Y > k\}} \right] \\
&\stackrel{\text{Lin.}}{=} \lim_{n \rightarrow \infty} \sum_{k=0}^n \mathbb{E}[\mathbf{1}_{\{Y > k\}}] \\
&\stackrel{3.3.2}{=} \sum_{k=0}^{\infty} \mathbb{P}(Y > k). \tag{b.2}
\end{aligned}$$

For the negative part Y^- , we have:

$$Y^- = \sum_{k=1}^{\infty} \mathbf{1}_{\{Y \leq -k\}} = \sum_{k=1}^{\infty} \mathbf{1}_{\{Y < -k+1\}}$$

We perform an index substitution by setting $j = -k + 1$. As k goes from 1 to $+\infty$, the new index j goes from 0 to $-\infty$:

$$Y^- = \sum_{j=-\infty}^0 \mathbf{1}_{\{Y < j\}}$$

For all $n \in \mathbb{N}$, let $Z_n = \sum_{j=-n}^0 \mathbf{1}_{\{Y < j\}}$. Since indicator functions are non-negative, $(Z_n)_{n \in \mathbb{N}}$ forms a monotonically increasing sequence of non-negative random variables that converges pointwise to Y^- , so that we can apply the Monotone Convergence Theorem again:

$$\begin{aligned}
\mathbb{E}[Y^-] &= \mathbb{E} \left[\lim_{n \rightarrow \infty} \sum_{j=-n}^0 \mathbf{1}_{\{Y < j\}} \right] \\
&\stackrel{\text{MCT}}{=} \lim_{n \rightarrow \infty} \mathbb{E} \left[\sum_{j=-n}^0 \mathbf{1}_{\{Y < j\}} \right] \\
&\stackrel{\text{Lin.}}{=} \lim_{n \rightarrow \infty} \sum_{j=-n}^0 \mathbb{E}[\mathbf{1}_{\{Y < j\}}] \\
&\stackrel{3.3.2}{=} \sum_{j=-\infty}^0 \mathbb{P}(Y < j). \tag{b.3}
\end{aligned}$$

Substituting the results from (b.2) and (b.3) into (b.1), we have:

$$\mathbb{E}[Y] = \sum_{k=0}^{\infty} \mathbb{P}(Y > k) - \sum_{k=-\infty}^0 \mathbb{P}(Y < k).$$

□

Part c)

We have that:

$$F_Z(z) = \mathbb{P}(Z \leq z) = \mathbb{P}(Z \in (-\infty, z]) = \mathbb{P}_Z((-\infty, z]) = \int_{\mathbb{R}} \mathbf{1}_{\{x \leq z\}} \mathbb{P}_Z(dx)$$

Substituting this definition of $F_Z(z)$ into the integral yields:

$$\int_{\mathbb{R}} F_Z(z) \mathbb{P}_Z(dz) = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} \mathbf{1}_{\{x \leq z\}} \mathbb{P}_Z(dx) \right) \mathbb{P}_Z(dz)$$

Since the indicator function is non-negative and measurable, and $\mathbb{P}_Z$ is a probability measure (and hence σ -finite), Tonelli's Theorem allows us to interchange the order of integration. The integration domain is $\{(x, z) \in \mathbb{R}^2 \mid x \leq z\}$, which means for a fixed x , the bounds for z are $[x, \infty)$. Reordering the integrals gives:

$$\begin{aligned} \int_{\mathbb{R}} \left(\int_{\mathbb{R}} \mathbf{1}_{\{x \leq z\}} \mathbb{P}_Z(dx) \right) \mathbb{P}_Z(dz) &= \int_{\mathbb{R}} \int_{\mathbb{R}} \mathbf{1}_{\{z \geq x\}} \mathbb{P}_Z(dz) \mathbb{P}_Z(dx) \\ &= \int_{\mathbb{R}} \left(\int_{[x, \infty)} 1 \mathbb{P}_Z(dz) \right) \mathbb{P}_Z(dx) \\ &= \int_{\mathbb{R}} \mathbb{P}_Z([x, \infty)) \mathbb{P}_Z(dx) \\ &= \int_{\mathbb{R}} \mathbb{P}(Z \in [x, \infty)) \mathbb{P}_Z(dx) \\ &= \int_{\mathbb{R}} \mathbb{P}(Z \geq x) \mathbb{P}_Z(dx) \end{aligned}$$

According to Lemma 1.4.2, the probability of a single point is $\mathbb{P}(Z = x) = F_Z(x) - F_Z(x-)$. Because F_Z is given as a continuous distribution function, $\mathbb{P}(Z = x) = 0$ for all $x \in \mathbb{R}$. Using this and the complement rule, we have that:

$$\mathbb{P}(Z \geq x) = \mathbb{P}(Z > x) = 1 - \mathbb{P}(Z \leq x) = 1 - F_Z(x).$$

Substituting this back into the integral:

$$\begin{aligned} \int_{\mathbb{R}} \mathbb{P}(Z \geq x) \mathbb{P}_Z(dx) &= \int_{\mathbb{R}} (1 - F_Z(x)) \mathbb{P}_Z(dx) \\ &= \int_{\mathbb{R}} 1 \mathbb{P}_Z(dx) - \int_{\mathbb{R}} F_Z(x) \mathbb{P}_Z(dx) \end{aligned}$$

The first integral in the expression evaluates to 1 since $\mathbb{P}_Z$ is a normalized probability measure (Definition 1.1.9). The second integral is exactly our initial integral, using x as the dummy variable instead of z . Since $0 \leq F_Z(z) \leq 1$ and $\mathbb{P}_Z(\mathbb{R}) = 1$, the integral $\int_{\mathbb{R}} F_Z(z) \mathbb{P}_Z(dz)$ is finite, which allows us to subtract it from both sides. Substituting this back into the equation:

$$\begin{aligned} \int_{\mathbb{R}} F_Z(z) \mathbb{P}_Z(dz) &= 1 - \int_{\mathbb{R}} F_Z(z) \mathbb{P}_Z(dz) \\ \iff 2 \int_{\mathbb{R}} F_Z(z) \mathbb{P}_Z(dz) &= 1 \\ \iff \int_{\mathbb{R}} F_Z(z) \mathbb{P}_Z(dz) &= \frac{1}{2} \end{aligned}$$

□